

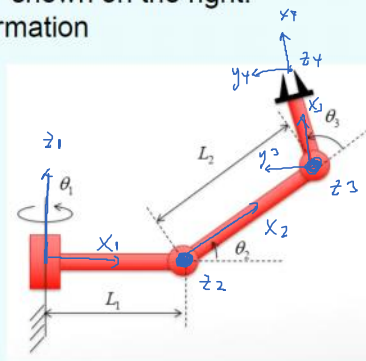
Question 1

- Find the Jacobian of the manipulator shown on the right. (You should already have some information about this robot in earlier tutorials).

- Write it in terms of frame {3} at the wrist of robot.

- Velocity propagation method. ✓
- Differentiation of kinematic equations. ✓

- Write also in terms of frame {4} at the tip of hand, having same orientation as {3}.



$${}^1P_2 = [L_1 \ 0 \ 0]^T$$

$${}^2P_3 = [L_2 \ 0 \ 0]^T$$

For this robot, we had (from forward kinematics tutorial)

$${}^0T_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^1T_2 = \begin{bmatrix} c_2 & -s_2 & 0 & L_1 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} c_3 & -s_3 & 0 & L_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^0T_3 = \begin{bmatrix} c_1 c_2 & -c_1 s_2 & s_1 & L_2 c_1 c_2 + L_1 c_1 \\ s_1 c_2 & -s_1 s_2 & -c_1 & L_2 s_1 c_2 + L_1 s_1 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A) velocity propagation $\rightarrow$ long. Useful for both linear & rotary Jacobian.

Start with ${}^0W_0 = 0$, ${}^0V_0 = 0$

$$\text{Joint 1: Revolute: } {}^1W_1 = \underbrace{{}^0R \cdot {}^0W_0}_{=0} + \underbrace{\hat{z}_1}_{\text{scalar}} \dot{\theta}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$${}^1V_1 = \underbrace{{}^0R \cdot {}^0V_0}_{=0} + \underbrace{{}^0W_0 \times {}^0P_1}_{=0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Joint 2: Revolute: } {}^2W_2 = \underbrace{{}^1R \cdot {}^1W_1}_{\underbrace{{}^1R^T}} + \hat{z}_2 \dot{\theta}_2$$

$$= \begin{bmatrix} c_2 & 0 & s_2 \\ -s_2 & 0 & c_2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix}$$

$$\begin{aligned}
 &= \begin{bmatrix} C_2 & 0 & S_2 \\ -S_2 & 0 & C_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix} \\
 &= \begin{bmatrix} S_2 \dot{\theta}_1 \\ C_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \quad \leftarrow
 \end{aligned}$$

$${}^2V_2 = \underbrace{{}^2R}^{{}^2R^T} (\underbrace{{}^1V_1}_{=0} + {}^1W_1 \times {}^1P_2)$$

$$\begin{aligned}
 &= \begin{bmatrix} C_2 & 0 & S_2 \\ -S_2 & 0 & C_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ L_1 & 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} C_2 & 0 & S_2 \\ -S_2 & 0 & C_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ L_1 \dot{\theta}_1 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ 0 \\ -L_1 \dot{\theta}_1 \end{bmatrix}
 \end{aligned}$$

Joint 3 : Revolute joint

$${}^3W_3 = \underbrace{{}^3R}^{{}^3R^T} {}^2W_2 + \dot{\theta}_3 {}^3\hat{z}_3$$

$$\begin{aligned}
 &= \begin{bmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_2 \dot{\theta}_1 \\ C_2 \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix} \\
 &= \begin{bmatrix} S_2 C_3 \dot{\theta}_1 + C_2 S_3 \dot{\theta}_1 \\ -S_2 S_3 \dot{\theta}_1 + C_2 C_3 \dot{\theta}_1 \\ \dot{\theta}_2 + \dot{\theta}_3 \end{bmatrix} \\
 &= \begin{bmatrix} S_{23} \dot{\theta}_1 \\ C_{23} \dot{\theta}_1 \\ \dot{\theta}_2 + \dot{\theta}_3 \end{bmatrix}
 \end{aligned}$$

$${}^3V_3 = \underbrace{{}^3R}^{{}^3R^T} ({}^2V_2 + {}^2W_2 \times {}^2P_3)$$

$$\begin{aligned}
 &= \begin{bmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 0 \\ 0 \\ -L_1 \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} i & j & k \\ S_2 \dot{\theta}_1 & C_2 \dot{\theta}_1 & \dot{\theta}_2 \\ L_2 & 0 & 0 \end{bmatrix} \right\} \\
 &= \begin{bmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 &= \begin{bmatrix} C_3 & S_3 & 0 \\ -S_3 & C_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 0 \\ 0 \\ -L_1 \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ L_2 \dot{\theta}_2 \\ -L_2 C_2 \dot{\theta}_1 \end{bmatrix} \right\} \\
 &= \begin{bmatrix} L_2 S_3 \dot{\theta}_2 \\ L_2 C_3 \dot{\theta}_2 \\ -L_1 \dot{\theta}_1 - L_2 C_2 \dot{\theta}_1 \end{bmatrix}
 \end{aligned}$$

Transfer back to $\{0\}$ frame:

$$\begin{aligned}
 {}^0W_3 &= {}^0_3R {}^3W_3 \\
 &= \begin{bmatrix} C_1 C_{23} & -C_1 S_{23} & S_1 \\ S_1 C_{23} & -S_1 S_{23} & -C_1 \\ S_{23} & C_{23} & 0 \end{bmatrix} \begin{bmatrix} S_{23} \dot{\theta}_1 \\ C_{23} \dot{\theta}_1 \\ \dot{\theta}_2 + \dot{\theta}_3 \end{bmatrix} \\
 &= \begin{bmatrix} C_1 C_{23} S_{23} \dot{\theta}_1 - C_1 S_{23} C_{23} \dot{\theta}_1 + S_1 (\dot{\theta}_2 + \dot{\theta}_3) \\ S_1 C_{23} S_{23} \dot{\theta}_1 - S_1 S_{23} C_{23} \dot{\theta}_1 - C_1 (\dot{\theta}_2 + \dot{\theta}_3) \\ S_{23} \dot{\theta}_1 + C_{23} \dot{\theta}_2 \end{bmatrix} \\
 &= \begin{bmatrix} S_1 (\dot{\theta}_2 + \dot{\theta}_3) \\ -C_1 (\dot{\theta}_2 + \dot{\theta}_3) \\ \dot{\theta}_1 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 {}^0U_3 &= {}^0_3R {}^3U_3 \\
 &= \begin{bmatrix} C_{23} & -C_1 S_{23} & S_1 \\ S_1 C_{23} & -S_1 S_{23} & -C_1 \\ S_{23} & C_{23} & 0 \end{bmatrix} \begin{bmatrix} L_2 S_3 \dot{\theta}_2 \\ L_2 C_3 \dot{\theta}_2 \\ -L_1 \dot{\theta}_1 - L_2 C_2 \dot{\theta}_1 \end{bmatrix} \\
 &= \begin{bmatrix} L_2 C_1 C_{23} S_3 \dot{\theta}_2 - L_2 C_1 S_{23} C_3 \dot{\theta}_2 - L_1 S_1 \dot{\theta}_1 - L_2 S_1 C_2 \dot{\theta}_1 \\ L_2 S_1 C_{23} S_3 \dot{\theta}_2 - L_2 S_1 S_{23} C_3 \dot{\theta}_2 + L_1 C_1 \dot{\theta}_1 + L_2 C_1 C_2 \dot{\theta}_1 \\ L_2 S_{23} S_3 \dot{\theta}_2 + L_2 C_{23} C_3 \dot{\theta}_2 \end{bmatrix}
 \end{aligned}$$

can be simplified further:

$$\begin{aligned}
 \text{row 1 \& 2} \quad & C_{23} S_3 - S_{23} C_3 \\
 &= - \left[\underbrace{S_{23} C_3}_{\tilde{A} \tilde{B}} - C_{23} S_3 \right] \quad \begin{matrix} \downarrow \\ S(A-B) = SA \tilde{C} B - C A S B \end{matrix} \\
 &= - \underbrace{S (\theta_2 + \theta_3)}_{\tilde{A}} - \underbrace{\theta_3}_{\tilde{B}} = \underline{\underline{-S_2}}
 \end{aligned}$$

$$\begin{aligned}
 \text{row 3} \quad & S_{23} S_3 + C_{23} C_3 \\
 &= \underbrace{C (\theta_2 + \theta_3)}_{\tilde{A}} - \underbrace{\theta_3}_{\tilde{B}} = \underline{\underline{C_2}}
 \end{aligned}$$

$$\downarrow = \frac{c}{A} (\theta_2 + \theta_3) - \frac{\theta_3}{B} = \underline{\underline{c_2}}$$

$${}^0V_3 = \begin{bmatrix} -L_2 c_1 s_2 \dot{\theta}_2 - L_1 s_1 \dot{\theta}_1 - L_2 s_1 c_2 \dot{\theta}_1 \\ -L_2 s_1 s_2 \dot{\theta}_2 + L_1 c_1 \dot{\theta}_1 + L_2 c_1 c_2 \dot{\theta}_1 \\ L_2 c_2 \dot{\theta}_2 \end{bmatrix}$$

Jacobian

$$\dot{x} = J \dot{\theta}$$

$${}^0 \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} -L_1 s_1 - L_2 s_1 c_2 & -L_2 c_1 s_2 & 0 \\ L_1 c_1 + L_2 c_1 c_2 & -L_2 s_1 s_2 & 0 \\ 0 & L_2 c_2 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$\uparrow {}^0J_V$

$${}^0W_3 = \begin{bmatrix} s_1(\dot{\theta}_2 + \dot{\theta}_3) \\ c_1(\dot{\theta}_2 + \dot{\theta}_3) \\ \dot{\theta}_1 \end{bmatrix} \Rightarrow {}^0 \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = \begin{bmatrix} 0 & s_1 & s_1 \\ 0 & c_1 & c_1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$\uparrow {}^0J_W$

end of part 1 !

B) Direct differentiation (linear Jacobian only)

Position of wrist = 4th column of 0_3T

$${}^0 \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\text{wrist}} = \begin{bmatrix} L_2 c_1 c_2 + L_1 c_1 \\ L_2 s_1 c_2 + L_1 s_1 \\ L_2 s_2 \end{bmatrix} = \begin{bmatrix} f_x(q) \\ f_y(q) \\ f_z(q) \end{bmatrix}$$

$q \text{ or } d$

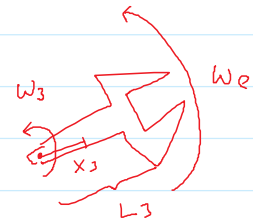
$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_x}{\partial q_1} & \frac{\partial f_x}{\partial q_2} & \frac{\partial f_x}{\partial q_3} \\ \frac{\partial f_y}{\partial q_1} & \frac{\partial f_y}{\partial q_2} & \frac{\partial f_y}{\partial q_3} \\ \frac{\partial f_z}{\partial q_1} & \frac{\partial f_z}{\partial q_2} & \frac{\partial f_z}{\partial q_3} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix}$$

$$= \begin{bmatrix} -L_2 s_1 c_2 - L_1 s_1 & -L_2 c_1 s_2 & 0 \\ L_2 c_1 c_2 + L_1 c_1 & -L_2 s_1 s_2 & 0 \\ 0 & L_2 c_2 & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix}$$

$\uparrow {}^0J_V$

Vel at tip of hand

$$\underline{w_e = w_3}$$



$${}^3V_e = {}^3V_3 + {}^3w_3 \times {}^3P_e$$

$$= \begin{bmatrix} L_2 s_3 \dot{\theta}_2 \\ L_2 c_3 \dot{\theta}_2 \\ -L_1 \dot{\theta}_1 - L_2 c_2 \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} i & j & k \\ s_{13} \dot{\theta}_1 & c_{23} \dot{\theta}_1 & \dot{\theta}_2 + \dot{\theta}_3 \\ L_3 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} L_2 s_3 \dot{\theta}_2 \\ L_2 c_3 \dot{\theta}_2 + L_3 (\dot{\theta}_2 + \dot{\theta}_3) \\ -L_1 \dot{\theta}_1 - L_2 c_2 \dot{\theta}_1 - L_3 c_2 \dot{\theta}_1 \end{bmatrix}$$

Transform back to $\{0\}$

$${}^0V_e = {}^0R_3 {}^3V_e$$

$$= \begin{bmatrix} -L_1 s_1 - L_2 s_1 c_2 - L_3 s_1 c_{23} & -L_2 c_1 s_2 - L_3 c_1 s_{23} & -L_3 c_1 c_{23} \\ L_1 c_1 + L_2 c_1 c_2 + L_3 c_1 c_{23} & -L_2 s_1 s_2 - L_3 s_1 s_{23} & -L_3 s_1 s_{23} \\ 0 & L_2 c_2 + L_3 c_{23} & L_3 c_{23} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

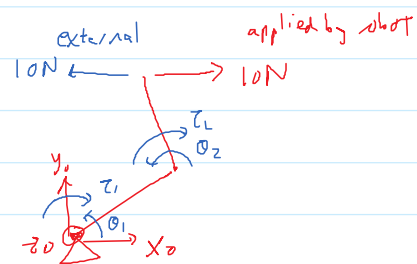
Question 2

- A 2-link manipulator has the following Jacobian:

$${}^0J_v = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}$$

- If we ignore gravity, what are the joint torques required so that the manipulator can apply a static force of

$$\underline{{}^0F = 10\hat{X}_0}$$



$$\underline{\tau = J^T F}$$

$$= \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & L_1 c_1 + L_2 c_{12} \\ -L_2 s_{12} & L_2 c_{12} \end{bmatrix} \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -10 L_1 s_1 - 10 L_2 s_{12} \\ -10 L_2 s_{12} \end{bmatrix}$$

↑
J is linear,
but time varying

J is linear,
but time varying
depending on θ